

Node-free modal Müller solver at high contrast, and its integration with shape continuation

Summary of findings, Neural-BEM, branch feature/shape-frequency-continuation

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Goal. Find why the node-free (Laurent) Müller solver fails on high-contrast scenes, fix it, and check whether it can replace the nodal physics inside the existing shape-continuation (SC) pipeline.

Conclusions at a glance

1. The high-contrast failure of the node-free assembler was a *precision* failure: the kernel was summed as a monomial series in $R = |z(\theta) - z(\varphi)|^2$ whose terms reach $I_0(kD) \approx 10^{23}$ and must cancel. Expanding the same functions in Chebyshev polynomials of R removes this. The terms then sum to 1 (Lemma 2).
2. The same device applied to $\log|W|^2$ removes the star-shaped restriction, so SC's C shapes now assemble node-free.
3. A Parseval test on the Chebyshev arrays gives node-free bounds on $|W|^2$ and R , and doubles as a check that the curve is simple.
4. Inside SC's unchanged Levenberg–Marquardt loop, the node-free forward solver and the Hadamard Jacobian built from SC-035's velocity coefficients reproduce every accept/reject decision of three recorded stages. These include two $K = 192$ stages with 19 frequencies at contrast 13.3. Endpoints agree to $\leq 6 \times 10^{-13}$.
5. Not fixed, and not solver problems: the trace cutoff K_u must still grow at high contrast (96–128 at 2.5 GHz), and the final update band must still reach the measured frontier. Geometry-side nodes (SC-035 projection, validity checks) remain. The main open cost is the per-geometry setup of 9–25 s on one CPU core.

Assumptions.

- 2D scalar TMz (transverse magnetic, E along the cylinder axis), lossless media, equal permeability, one interface per object.
- Sand host with relative permittivity 6. Lengths are in package units of 5 cm; the 2.5 GHz exterior wavenumber is $k_e = 6.417$ per unit.
- Acquisition: SC's ring of 24 paired sources and receivers at radius 0.30 m, receiver offset 0.06 m, source strength 10^{-6} .
- “Given” next to a step marks a fact taken from the cited reference without proof.
- All timings are on one CPU core. The recorded runs used GPU nodal assembly, so their times are not comparable.

Notation

Symbol	Meaning
t, θ, φ	curve parameter; θ for the target point, φ for the source point
$z(t) = \sum_{ j \leq K} z_j e^{ijt}$	stored Cartesian curve as a complex number; the native code divides by $ z_1 $
K, M	half-band of the stored curve; half-band of the normal update
K_u, B	trace cutoff (modes $ m \leq K_u$); coefficient window of the kernel arrays
k_e, k_i, χ	exterior and interior wavenumber; contrast $\chi = k_i^2/k_e^2$ (0.5 plastic, 13.3 water)
$w = e^{i\theta}, v = e^{i\varphi}$	unit-circle variables
R, r, D	$R = z(\theta) - z(\varphi) ^2$, $r = \sqrt{R}$, object diameter $D = \max r$
W	divided difference $W = (z(w) - z(v))/(w - v)$
G_k, P_k, Q_k	free-space Green's function $\frac{i}{4} H_0^{(1)}(kr)$; entire functions with $G_k = P_k(R) \log R + Q_k(R)$
σ_p	Taylor coefficients $(-k^2/4)^p/(p!)^2$ of $J_0(k\sqrt{R})$ in powers of R
T_n	Chebyshev polynomial, $T_n(\cos x) = \cos nx$, so $ T_n \leq 1$ on $[-1, 1]$
$R_{\max}, \ell$	upper bound of R ; $\ell = 2R/R_{\max} - 1 \in [-1, 1]$
c_n, d_n	Chebyshev coefficients of a radial function, and of $\log$
α, η_n	$\alpha = k\sqrt{R_{\max}}/2$; $\eta_0 = 1$, $\eta_n = 2$ for $n \geq 1$
β, Λ, λ	lower and upper bound of $ W ^2$; affine map of $ W ^2$ onto $[-1, 1]$
$N = -iz'$	unnormalised outward normal (only this meaning; node counts are written in words)
ν, ds	unit outward normal, arclength element
a_i, V_i	SC update coordinates (metres); velocity $V_i = \partial \mathcal{T}_z / \partial a_i$ of SC-035's trial curve
$u_s, \tilde{u}_r, Y_{rs}$	boundary trace from source s ; reciprocal trace from receiver r ; scattered data
ω_i	Hadamard weight $\text{Re}(V_i \bar{N})$
eps	double-precision unit roundoff, 1.1×10^{-16}
$I_0, J_n, H_n^{(1)}$	modified Bessel, Bessel and Hankel functions

1 Context: the MA and SC tracks

(read) Both tracks live on `feature/shape-frequency-continuation`, 230 commits ahead of `main`.

- **SC** (shape/frequency continuation, SC-001 to SC-051) is the inverse pipeline. It fits Fourier curves by Levenberg–Marquardt (LM) over a frequency ladder, with nodal Müller/Kress physics at 512 nodes and a 1024-node acceptance check.
- **MA** (modal atlas, MA-001 to MA-006) studies high contrast. Its best policy, **DF** (MA-005), runs as follows:
 - Mie-based localisation on damped data;
 - a damped prefix at $k(1 + 0.25i)$ with $K = 4 \rightarrow 20$;
 - release and fixed stages at $K = 192$ over all 19 catalog frequencies (0.25–2.5 GHz);
 - a final band grown to the measured frontier (M up to 67 for the replayed scene).
- The node-free code, `experiments/modal_muller_research/`, is unchanged on this branch. MA-006 used a projected-Nyström control and states that no native assembly was qualified.

The concern about the modal Müller solver at high contrast splits into four different problems. Only the first two belong to the native solver; the last two are physics and policy that a nodal solver pays too.

#	Issue	Source	Native only?
1	power series in R loses precision at large kD	package README (“broken at 5 GHz”)	yes
2	half-plane log certificate fails for non-star curves	package README; SC’s C and kite	yes
3	K_u must grow; reduced solves need more than projection	MA-006 (48 $\rightarrow$ 64–96 at $\chi = 13.3$)	no
4	final band must reach the frontier	MA-004/005 (up to 84)	no

2 Why the native assembler failed at high contrast

(*read*) The code writes the Green’s function in split form and sums each radial function as a series on coefficient arrays of R^p :

$$G_k = P_k(R) \log R + Q_k(R), \quad P_k(R) = -\frac{1}{4\pi} J_0(k\sqrt{R}) = -\frac{1}{4\pi} \sum_{p \geq 0} \sigma_p R^p. \quad (1)$$

The assembler needs six such entire functions of R : differences of P and Q , their derivatives, and k^2 -weighted versions.

Lemma 1 (size of the terms) For $0 \leq R \leq D^2$,

$$\begin{aligned} \sum_p |\sigma_p| R^p &\leq \sum_p \frac{(kD/2)^{2p}}{(p!)^2} & |\sigma_p| &= (k^2/4)^p / (p!)^2, \quad R \leq D^2 \\ &= I_0(kD) & & \text{series of } I_0 \text{ (given, DLMF §10.25)} \end{aligned} \quad (2)$$

The sum itself is $|J_0| \leq 1$, so the terms must cancel. The rounding error is therefore about $\text{eps } I_0(kD)$, using the larger of $k_e D$ and $k_i D$ (*derived estimate*).

Check (*ran*). The test object is the SC star truth $0.05(1 + 0.25 \cos 5t)e^{it}$ m at 2.5 GHz, with $D \approx 2.38$ units. Errors are relative Frobenius differences of the full Müller matrix against the nodal Kress oracle (1024 nodes, projected to modes, $K_u = 64$, $B = 160$).

χ	dominant kD	predicted $\text{eps } I_0(kD)$	measured (best of 28–120 terms)
0.5	$k_e D = 15.3$	5×10^{-11}	1.9×10^{-11}
4	$k_i D = 30.5$	1.4×10^{-4}	8.1×10^{-5}
13.3	$k_i D = 55.7$	9×10^6	5.5×10^6

More terms do not help, because the floor is rounding, not truncation. The failure comes from the monomial basis, not from avoiding nodes.

3 Fix 1: Chebyshev polynomials in R

(*derived; ran*) Bound R without nodes, map it to $[-1, 1]$, and build the polynomials by their recurrence on coefficient arrays:

$$R_{\max} = \left(2 \sum_{j \neq 0} |z_j|\right)^2 \geq \max R \quad |z(\theta) - z(\varphi)| \leq \sum_{j \neq 0} |z_j| |e^{ij\theta} - e^{ij\varphi}| \quad (3)$$

$$f(R) = \sum_{n \geq 0} c_n T_n(\ell(R)) \quad \ell = 2R/R_{\max} - 1 \in [-1, 1] \quad (4)$$

$$T_0 = 1, \quad T_1 = \ell, \quad T_{n+1} = 2\ell T_n - T_{n-1} \quad \text{three-term recurrence} \quad (5)$$

Each step of (5) is one exact polynomial product with ℓ , cut to the window B , as the code already did for R^p . The arrays $T_n(\ell(R))$ do not depend on k , so frequency-independent precomputation still works. The c_n come from a scalar calculation in the one variable R ; no boundary point is evaluated.

Lemma 2 (the Chebyshev terms stay small) Put $\alpha = k\sqrt{R_{\max}}/2$ and $R = R_{\max} \sin^2(\vartheta/2)$. Then

$$\begin{aligned}
J_0(2\alpha \sin \frac{\vartheta}{2}) &= J_0(\alpha)^2 + 2 \sum_{n \geq 1} J_n(\alpha)^2 \cos n\vartheta && \text{Graf, equal radii (given, DLMF §10.23)} \\
\ell = 2 \sin^2 \frac{\vartheta}{2} - 1 &= -\cos \vartheta, \quad T_n(\ell) = (-1)^n \cos n\vartheta && \text{by (4), } T_n(\cos x) = \cos nx \\
J_0(k\sqrt{R}) &= \sum_{n \geq 0} \eta_n (-1)^n J_n(\alpha)^2 T_n(\ell) && \text{substitute} \\
\sum_n |c_n| \sup |T_n| &= J_0(\alpha)^2 + 2 \sum_{n \geq 1} J_n(\alpha)^2 = 1 && |T_n| \leq 1; \text{ DLMF 10.23.3}
\end{aligned} \tag{6}$$

Compare (2): the monomial terms add up to $I_0(2\alpha)$, the Chebyshev terms to 1. Since $J_n(\alpha)$ is negligible once n is well past α , about $\alpha + O(\alpha^{1/3})$ terms are needed.

Check. At $\chi = 13.3$, $\alpha = 23.40 \cdot 2.5/2 = 29.25$. The Airy approximation (DLMF §10.19) gives $J_{30}(29.25) \approx 0.112$, so $|c_{30}| \approx 2(0.112)^2/4\pi = 2.0 \times 10^{-3}$. The computed coefficients are 2.0×10^{-3} at $n = 30$ and 5×10^{-9} at $n = 40$ (ran).

Result on the SC star (ran; same oracle as Section 2):

χ	monomial (best)	Chebyshev degree 40	degree 60	degree 80
0.5	1.9×10^{-11}	8.3×10^{-15}	8.3×10^{-15}	8.3×10^{-15}
4	8.1×10^{-5}	1.8×10^{-14}	1.8×10^{-14}	1.8×10^{-14}
13.3	5.5×10^6	9.5×10^{-8}	3.4×10^{-14}	3.4×10^{-14}

4 Fix 2: non-star curves, and node-free bounds

Problem (ran). The code needs $\text{Re}(W/z_1) > 0$ to use its series for $\log W$. On SC's C truth the sampled minimum is -1.21 (code bound -2.37), so it refuses. The kite passes by value ($+0.12$) but not by the coefficient bound.

Why only $|W|^2$ matters (read; derived). The assembler uses only $\log R$, and $|w - v|^2 = 4 \sin^2 \frac{\theta - \varphi}{2}$:

$$\log R = \log \left(4 \sin^2 \frac{\theta - \varphi}{2} \right) + \log |W|^2. \tag{7}$$

The first term has known Fourier coefficients. The second is smooth whenever $W \neq 0$, which holds exactly when the curve is regular and does not cross itself.

Fix (derived; ran). Apply the Chebyshev idea to $\log$ on $[\beta, \Lambda]$:

$$\begin{aligned}
\beta &\leq |W|^2 \leq \Lambda && \Lambda = \text{sum of } |\text{coefficients of } |W|^2| \text{ (triangle inequality)} \\
\lambda &= \frac{2|W|^2 - \Lambda - \beta}{\Lambda - \beta} \in [-1, 1] && \text{affine map} \\
\log |W|^2 &= \sum_n d_n T_n(\lambda) && \text{Chebyshev series of } \log \text{ on } [\beta, \Lambda] \\
|d_n| &\lesssim \rho^{-n}, \quad \rho = \frac{\sqrt{\Lambda} + \sqrt{\beta}}{\sqrt{\Lambda} - \sqrt{\beta}} && \text{Trefethen, ATAP Thm 8.2 (given)}
\end{aligned} \tag{8}$$

Check. On the C, $\Lambda = 7.43$ and $\beta = 0.05$ give $\rho = 2.950/2.502 = 1.179$. So $\ln 10^{16}/\ln \rho \approx 36.8/0.164 \approx 225$ terms are needed; the code used 230 (ran). A Taylor series in $|W|^2$ would need about 1400.

Choosing β without nodes (ran). If β is too large, $|\lambda| > 1$ somewhere and $T_n(\lambda)$ grows exponentially. By Parseval, the root-mean-square (RMS) of the coefficient array shows this. So halve β until the RMS stays at most 1. On the C, where the true minimum of $|W|^2$ is 0.197:

β	RMS of $T_n(\lambda)$ at $n = 50 / 100 / 200$
0.10	0.69 / 0.71 / 0.70
0.18	0.72 / 0.70 / 0.70
0.22	$5.0 / 1.0 \times 10^3 / 5.8 \times 10^7$
0.30	$4.3 \times 10^3 / 4.9 \times 10^8 / 8.9 \times 10^{18}$

This is an after-the-fact test, not a proof. It also works as a node-free check that the curve is simple, since no $\beta > 0$ passes if W vanishes somewhere.

Tighter $R_{\max}$ by the same test (*derived; ran*). If an upper bound $\tau < \max R$, then $\ell = 2R/\tau - 1 > 1$ somewhere and T_n grows. So bisect:

$$R_{\max}^* = \min \left\{ \tau : \max_{n \leq 60} \text{RMS } T_n(2R/\tau - 1) \leq 1 \right\}, \quad \text{then use } 1.02 R_{\max}^*. \quad (9)$$

curve	bound (3)	$\sum R \text{ coefficients} $	bound (9)	sampled max R
development C, $K = 10$	25.3	16.6	8.44	8.26
DF endpoint, $K = 192$	–	–	8.46	–

Result on the C truth (*ran; C rebuilt from the atlas_cases parameters, 2.5 GHz, $K_u = 64$, $B = 192$, relative matrix error against nodal Kress*):

reference	$\chi = 0.5$	$\chi = 13.3$
nodal, 1024 nodes	5.9×10^{-14}	1.7×10^{-13}
nodal, 2048 nodes	6.0×10^{-14}	1.8×10^{-13}

Window check: $B = 96$ gives 2.9×10^{-10} , and $B = 128$ or 160 give 1.7×10^{-13} . The star through the same general route gives 3.4×10^{-14} .

5 Supporting pieces

5.1 Scalar step in double precision

(*derived; ran*) The six radial functions can be evaluated at Chebyshev points of $[0, R_{\max}]$ in double precision from closed forms. The cancellation the Müller difference allows is done by hand; for example, for $f_3 = f_0/R + f'_1$:

$$\begin{aligned}
f_0/R &= -\frac{J_0(k_e r) - J_0(k_i r)}{4\pi R} & f_0 &= P_{k_e} - P_{k_i} \\
Q'_k(R) &= -\frac{i k H_1^{(1)}(kr)}{8r} - \frac{k J_1(kr) \log R}{8\pi r} + \frac{J_0(kr)}{4\pi R} & \text{differentiate } Q_k \text{ (below)} \\
f_3 &= \frac{1}{2r} \left[-\frac{i}{4} (k_e H_1^{(1)}(k_e r) - k_i H_1^{(1)}(k_i r)) \right. \\
&\quad \left. - \frac{\log R}{4\pi} (k_e J_1(k_e r) - k_i J_1(k_i r)) \right] \quad (10)
\end{aligned}$$

Here $Q_k = \frac{i}{4} H_0^{(1)}(kr) + \frac{1}{4\pi} J_0(kr) \log R$ and $dr/dR = 1/(2r)$. Adding f_0/R and $Q'_{k_e} - Q'_{k_i}$, the two $J_0/(4\pi R)$ terms cancel, which gives (10).

Taylor series are used only where $\max|k|r < 2$, where they are safe. The result matches a 60-digit computation to $\leq 1.4 \times 10^{-14}$ relative, for real and damped k , in 3–4 ms instead of 3.2–3.5 s (*ran*).

5.2 Damped wavenumbers

Lemma 3 (complex α)

$$\begin{aligned}
e^{i\alpha \sin \theta} &= \sum_{n \in \mathbb{Z}} J_n(\alpha) e^{in\theta} && \text{generating function (given, DLMF §10.12)} \\
\sum_n |J_n(\alpha)|^2 &= \frac{1}{2\pi} \int_0^{2\pi} e^{-2\operatorname{Im}(\alpha) \sin \theta} d\theta = I_0(2\operatorname{Im}\alpha) && \text{Parseval; } |e^{ix}| = e^{-\operatorname{Im}x}; \text{ DLMF §10.32} \\
\sum_{n \geq 0} |c_n| &= \sum_{n \geq 0} \eta_n |J_n(\alpha)|^2 = I_0(2|\operatorname{Im}\alpha|) && \text{Lemma 2 continued to complex } \alpha \quad (11)
\end{aligned}$$

For real α this gives 1. The expected rounding loss is $\epsilon I_0(2|\operatorname{Im}\alpha|)$, summed over the exterior and interior parts.

Check (*ran*; *C* truth, $\chi = 13.3$, $K_u = 64$, damped Kress at 2048 nodes as reference, loose $R_{\max}$ from (3)):

freq. (GHz)	$\operatorname{Im}\alpha_i$	predicted $\sum c_n $	measured	native data error
0.5	2.4	≈ 23	21	6.1×10^{-15}
1.0	4.75	1.73×10^3	1.77×10^3	3.2×10^{-13}
2.5	11.9	1.7×10^9	1.74×10^9	3.9×10^{-7}

DF damps only the prefix (up to 1 GHz), where fewer than 3 digits are lost. The tight bound (9) reduces α further; that case was not re-run.

5.3 Large curve bands: FFT arrays

(*read*; *ran*) The repo stores geometry polynomials as Python dicts. At $K = 192$, W has $K(K+1) = 37,056$ terms, so $W\overline{W}$ needs about 1.4×10^9 term pairs. Storing them as centred 2D arrays and multiplying by FFT fixes this. Cropping is exact:

Lemma 4 (window cropping) *Keep a product only for $|m| \leq b$, and let the first factor live in $|m| \leq b$. Then*

$$\begin{aligned}
(\xi\zeta)_m &= \sum_c \xi_{m-c} \zeta_c && \text{product of Fourier series} \\
|m| \leq b, |m-c| \leq b &\Rightarrow |c| \leq 2b && \text{triangle inequality} \quad (12)
\end{aligned}$$

so the second factor can be cut to $|c| \leq 2b$ without changing the kept window (per axis).

On the $K = 10$ C, the array version reproduces the dict version: 1.37×10^{-8} at $K_u = 64$ and 5.7×10^{-13} at $K_u = 96$ (*ran*).

6 Integration with shape continuation

6.1 Where SC uses nodes

(*read*: *updates.py*, *geometry.py*, *forward.py*, *lm.backend.py*, *SC-035 state.update.py*)

SC step	now	node-free replacement	status
forward solve	Kress, 512 nodes; 1024-node acceptance check	Sections 3–5	ran
sources, receivers	nodal	Graf expansions (<code>RegularWaves</code>); k_e only	exists
Jacobian	Hadamard on nodal traces	(14) with SC-035's V_i	ran
update and refit	SC-035: $\mathcal{T}_z(a) = z + \Pi_K[\mathcal{A}(z + h\nu) - \mathcal{A}(z)]$ with nodes and splines	modal speed control, weighted lift	derived only
validity	sampled self-intersection; resolution gate	Parseval test; K_u and B doubling	partly
frontier, band M	column norms of J	same, from the coefficient J	direct

Here $\mathcal{A}$ is SC's arclength reparameterisation, Π_K truncation to band K , and h the normal displacement.

6.2 The Jacobian from SC-035's velocities

SC-035's `space.derivatives` is exactly V_i as Cartesian coefficient arrays of shape $(2K + 1) \times (\text{number of } a_i)$. So the physics can take V_i as coefficients directly.

Lemma 5 (normal velocity in the stored parameter) *For a counterclockwise curve,*

$$\begin{aligned}
N &= -iz' = |z'|\nu & \text{rotating the tangent by } -90^\circ \\
(V \cdot \nu) ds &= \text{Re}(V\bar{\nu})|z'| dt = \text{Re}(V\bar{N}) dt & ds = |z'| dt, \bar{N} = |z'|\bar{\nu}
\end{aligned} \tag{13}$$

With $N_j = jz_j$, the Hadamard derivative becomes a finite contraction of coefficient arrays:

$$\begin{aligned}
\omega_i &= \frac{1}{2}(V_i\bar{N} + \bar{V}_iN) & \text{Re } x &= (x + \bar{x})/2; \text{ band } 2K \\
\partial_{a_i} Y_{rs} &= (k_i^2 - k_e^2) \int_0^{2\pi} u_s \tilde{u}_r \omega_i dt = 2\pi(k_i^2 - k_e^2) \sum_m \omega_{i,m}(u_s \tilde{u}_r)_{-m} & \text{Hadamard, by (13)}
\end{aligned} \tag{14}$$

6.3 How the native physics was plugged in

Only two functions were replaced: `lm.backend.solve` (native forward) and `Objective.jacobian` (native (14)). Everything else was the recorded configuration, data and SC-035 update. The production and refined node counts 512 and 1024 were mapped to $K_u = 64$ and 96 for $K \leq 24$ stages, and to 96 and 128 for $K = 192$ stages.

7 Verification

7.1 Forward data on SC's acquisition

(*ran*) The curve is the saved C truth SC-050/inputs/development_c ($K = 10$). Errors are relative to SC's `forward.solve` at 2048 nodes; SC's own 512- versus 2048-node difference is $\leq 2.1 \times 10^{-14}$.

freq. (GHz)	χ	$K_u = 32$	$K_u = 48$	$K_u = 64$
0.5	0.5	2.8×10^{-14}	3.7×10^{-15}	3.8×10^{-15}
0.5	13.3	2.6×10^{-10}	5.1×10^{-14}	5.3×10^{-14}
1.0	0.5	2.8×10^{-12}	6.5×10^{-15}	6.2×10^{-15}
1.0	13.3	1.4×10^{-6}	4.6×10^{-12}	1.2×10^{-13}
2.5	0.5	4.0×10^{-8}	3.3×10^{-13}	3.3×10^{-13}
2.5	13.3	8.7×10^{-2}	1.9×10^{-4}	1.4×10^{-8}

At $K = 192$ (*ran*; DF endpoint of *shifted_rotated_c*, $\chi = 13.3$, relative to 2048 nodes):

freq. (GHz)	nodal 1024	$B=128, K_u=64$	$B=128, K_u=96$	$B=160, K_u=128$
0.25	9×10^{-16} (5.2 s)	1.7×10^{-10}	9.2×10^{-15} (1.5 s)	9.3×10^{-15}
1.25	5×10^{-15} (7.3 s)	5.9×10^{-9}	2.1×10^{-13} (1.2 s)	1.9×10^{-13}
2.5	2×10^{-14} (6.8 s)	6.8×10^{-7}	1.6×10^{-10} (1.5 s)	1.2×10^{-12} (1.5 s)

The only remaining error is the trace cutoff K_u , as MA-006 predicts.

7.2 Jacobian (14) against SC's Jacobian

(*ran*) All at $\chi = 13.3$. Entries are relative Frobenius errors, with the worst single column in brackets.

curve, band	GHz	ref. nodes	$K_u = 64$	$K_u = 96$	$K_u = 128$
C, $K = 24, M = 12$	1.0	1024	1.9×10^{-13}	2.0×10^{-13}	–
C, $K = 24, M = 24$	1.0	1024	1.9×10^{-13}	1.9×10^{-13}	–
C, $K = 24, M = 12$	2.5	1024	2.8×10^{-8}	1.3×10^{-12}	–
C, $K = 24, M = 24$	2.5	1024	9.1×10^{-8}	1.3×10^{-12}	–
DF, $K = 192, M = 67$	1.25	2048	–	6.6×10^{-11} (4.6×10^{-6})	4.6×10^{-13} (3.7×10^{-9})
DF, $K = 192, M = 67$	2.5	2048	–	8.7×10^{-8} (1.3×10^{-5})	5.6×10^{-10} (5.3×10^{-8})

The worst columns belong to the highest update harmonics, which are the smallest columns.

7.3 Replays of recorded stages

(*ran*) All three are from scene *shifted_rotated_c* at $\chi = 13.3$.

stage	recorded (nodal, GPU)	native replay
D stage_2_damped ($K = 12, M = 5$, two damped frequencies)	10 accepted of 25 trials; final loss 0.004376567107229278	all 25 trial statuses identical; final 0.004376567107229138. A nodal replay here was also identical (final ... 229094).
D release_M11 ($K = 192, M = 11, 19$ frequencies)	7 of 7 accepted; 0.11386 $\rightarrow$ 0.009759024799226725	same 7 statuses; final 0.009759024799275914 (5×10^{-12} rel.); endpoint differs by 5.7×10^{-13} units (3×10^{-14} m); 304 solves, as recorded
DF fixed_M43 ($K = 192, M = 43, 19$ frequencies)	1 accepted; $1.3292 \times 10^{-9} \rightarrow 2.4723566206 \times 10^{-11}$	same status; final $2.4723565413 \times 10^{-11}$ (3×10^{-8} rel.); endpoint differs by 2×10^{-14} units

7.4 Cost

(*ran*; one CPU core)

item	time
nodal Kress assembly + LU, $K = 10$ C: 256 / 512 / 1024 nodes	0.27 / 1.10 / 4.40 s
nodal solve, $K = 192$ curve, 1024 nodes	5.2–7.5 s per frequency
native solve, $K = 192, B = 128, K_u$ 96–128	1.0–1.7 s per frequency
native geometry setup per new curve ($\log W ^2$, bound (9), T_n arrays)	9–25 s
native Jacobian contraction (14), all frequencies of a stage	0.2–0.7 s
stage_2_damped : nodal replay / native replay (sharing the core)	809 s / 634 s
fixed_M43 : native replay (geometry 24 s, 76 solves 86 s)	116 s
release_M11 : native replay (geometry 105 s, 304 solves 315 s)	425 s
recorded GPU nodal runs: stage_2_damped / release_M11 / fixed_M43	115 / 22 / 9 s

8 Conclusions

What is fixed.

1. **High-contrast precision.** The monomial series cancelled terms of size $I_0(kD)$, which is 10^{23} at $\chi = 13.3$ and 2.5 GHz. Chebyshev expansion in R bounds the term sum by 1 (Lemma 2). Matrices then match nodal Kress to $\leq 2 \times 10^{-13}$ at χ up to 13.3.
2. **Non-star shapes.** Chebyshev expansion of $\log|W|^2$ replaces the half-plane certificate. SC's C shapes assemble to $\leq 2 \times 10^{-13}$.
3. **Node-free bounds.** The Parseval test (9) and its lower-bound version give β and a tight $R_{\max}$ without sampling the boundary, and signal a self-intersecting curve.
4. **Damped prefix.** Complex wavenumbers work; the loss is bounded by $I_0(2|\text{Im}\alpha|)$ (Lemma 3) and stays under 3 digits in DF's prefix.
5. **Large K .** FFT arrays with exact cropping (Lemma 4) make $K = 192$ practical.
6. **Integration.** SC-035 already provides V_i as coefficients, so the Jacobian (14) is node-free. Swapping only the physics into SC's unchanged LM loop reproduced every decision of three recorded stages, including two $K = 192$, 19-frequency stages at $\chi = 13.3$.

What is not fixed.

- **Trace cutoff.** K_u must still grow with contrast and frequency: 64 is enough for the $K = 10$ C to 10^{-8} , and the $K = 192$ endpoint needs 96 (production) and 128 (refined) to pass the 10^{-7} resolution check at 2.5 GHz. This is physics and applies to any solver.
- **Update band.** The final band M must still reach the measured frontier (66 for the replayed scene, up to 84 in MA-004). This is policy.
- **Geometry-side nodes.** SC-035's projection and the validity checks still use nodes and splines. The node-free replacements (modal speed control, weighted lift) are derived but not implemented.
- **Cost.** The per-geometry setup (9–25 s on one core) is the bottleneck. Per-frequency solves are already cheaper than 1024-node nodal solves on the same core, but nothing is ported to GPU.

Suggested next steps, in order.

1. Package the code as a new `experiments/laurent_chebyshev/` backend selectable next to `SC_FORWARD_BACKEND`, without editing hash-pinned files.
2. Cut the per-geometry cost: warm-start the bound (9) as β already is; reuse T_n arrays between nearby trial curves; port the FFT products to GPU.
3. Replace the node-count pair (512, 1024) by a K_u pair chosen per frequency, using MA-006's measured cutoffs.
4. Test the remaining cases: damped solves with the tight $R_{\max}$, multi-object scenes, other scenes and contrasts, and one full DF run from localisation to the last fixed stage.
5. Only then replace the geometry side. This changes what M means (harmonics in t rather than arclength), so SC-035, SC-050 and DF thresholds would need to be measured again.

Remarks

1. **Stability of the cut recurrence** (*derived*). With truncation at B , (5) computes $T_n(A)\mathbf{1}$ with A the multiplication by ℓ restricted to the window. A is self-adjoint with eigenvalues in $[-1, 1]$. So rounding errors grow at most linearly in n .

2. **Hadamard versus the discrete model.** At small K_u the Hadamard formula on cut traces differs from the derivative of the cut discrete model (MA-006). In the replays, K_u was large enough that this did not change any LM decision.
3. **Corrections made during the session.** N was briefly used for both the unnormalised normal and the node count; here it is only the normal. Early timings labelled “8 threads” were in fact on one core.
4. **Not read:** SC’s `validity.py` and `policy.py`, MA’s `frontier_tail.py`, and SC-050’s `run.py`. Nothing above depends on them beyond what the replays exercised.
5. **Prior art.** Chebyshev polynomials instead of monomials is standard numerical analysis; I did not search for its use inside a Fourier–Galerkin boundary integral assembly.
6. **Code.** Seven files (about 450 lines) wrap or subclass repo classes: `cheb_operator.py`, `fast_scalars.py`, `native_sc.py`, `array_native.py`, and three replay scripts. Run them from the repository root with `PYTHONPATH=solvers:.<folder>`.

Result. Expanding the kernel in Chebyshev polynomials of R and $|W|^2$ makes the node-free Müller solver accurate at contrast 13.3, for non-star shapes and at $K = 192$. Plugged into SC’s unchanged LM loop through SC-035’s velocity coefficients, it reproduces recorded high-contrast stages to $\leq 6 \times 10^{-13}$. What remains is speed and the geometry side, not correctness.

Intuition. Powers of R become huge across the object and must cancel, while Chebyshev polynomials of R never exceed 1 there, so the same kernel is built without large intermediate numbers.